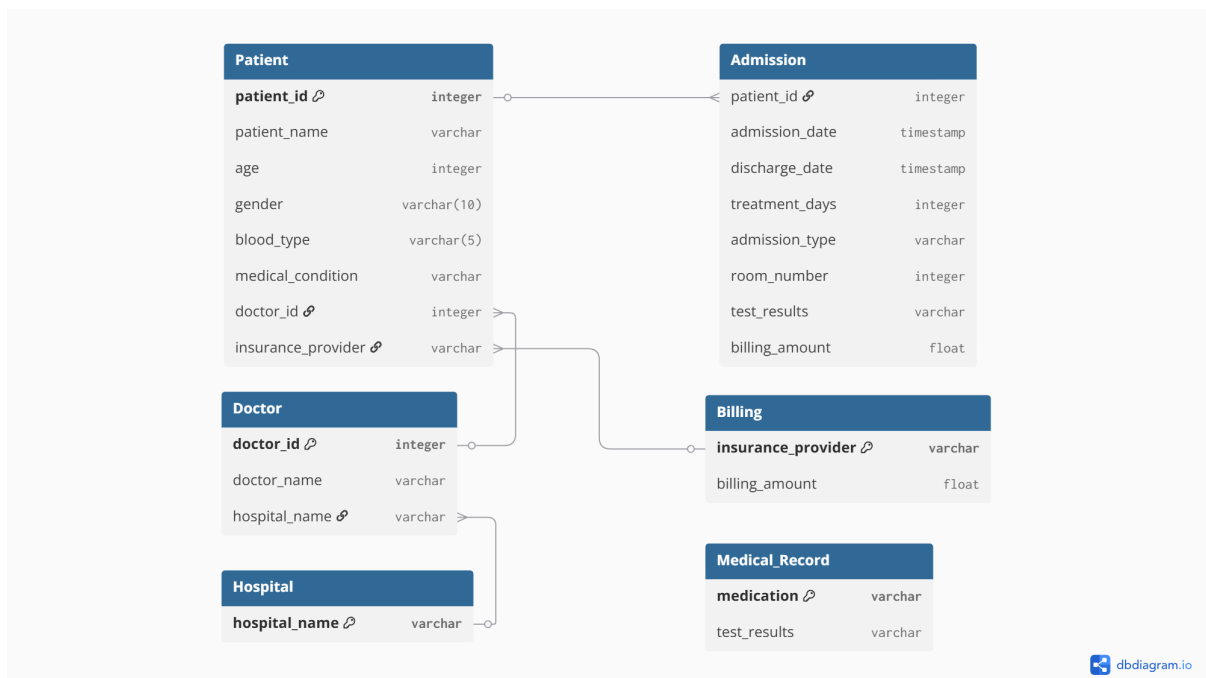
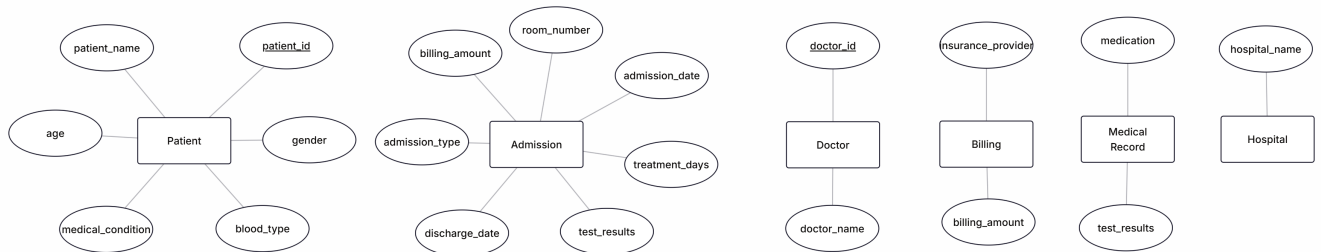


Healthcare SQL Data Analysis

John Doe

Diagrams:

In order to work with SQL, relational diagrams must be made to ensure clarity with the attributes to see what tables must be created and what relations a table can have with another. The following tables were created:



With this visualization, 6 tables were created for organization and to create relationships one another.

SQL Tables Created:

```
1 CREATE TABLE Hospital (  
2     hospital_name VARCHAR(150) PRIMARY KEY  
3 );  
4  
5 CREATE TABLE Doctor (  
6     doctor_id INT PRIMARY KEY,  
7     doctor_name VARCHAR(100),  
8     hospital_name VARCHAR(150),
```

```

9      CONSTRAINT FK_Doctor_Hospital FOREIGN KEY (hospital_name)
      REFERENCES Hospital(hospital_name)
10 );
11
12 CREATE TABLE Billing (
13     insurance_provider VARCHAR(100) PRIMARY KEY,
14     billing_amount DECIMAL(10,2)
15 );
16
17 CREATE TABLE Patient (
18     patient_id INT PRIMARY KEY,
19     patient_name VARCHAR(100),
20     age INT,
21     gender VARCHAR(10),
22     blood_type CHAR(3),
23     medical_condition VARCHAR(100),
24     doctor_id INT,
25     insurance_provider VARCHAR(100),
26     CONSTRAINT FK_Patient_Doctor FOREIGN KEY (doctor_id)
      REFERENCES Doctor(doctor_id),
27     CONSTRAINT FK_Patient_Billing FOREIGN KEY
      (insurance_provider) REFERENCES Billing(insurance_provider)
28 );
29
30 CREATE TABLE Admission (
31     admission_id INT IDENTITY(1,1) PRIMARY KEY,
32     patient_id INT,
33     admission_date DATE,
34     discharge_date DATE,
35     treatment_days INT,
36     admission_type VARCHAR(20),
37     room_number INT,
38     test_results VARCHAR(20),
39     billing_amount DECIMAL(10,2),
40     CONSTRAINT FK_Admission_Patient FOREIGN KEY (patient_id)
      REFERENCES Patient(patient_id)
41 );
42
43 CREATE TABLE Medical_Record (
44     medication VARCHAR(100) PRIMARY KEY,
45     test_results VARCHAR(20)
46 );

```

Listing 1: Table Creation

SQL Queries:

Patient Counts:

```

1 -- Query for Patient Count:

```

```
2 SELECT COUNT(*) AS total_patients FROM Patient;
```

Gender Counts:

```
1 -- Query for Gender Count:
2 SELECT gender, COUNT(*)
3 FROM Patient
4 GROUP BY gender;
```

Distinct Medical Conditions:

```
1 -- Distinct Medical Conditions:
2 SELECT DISTINCT medical_condition
3 FROM Patient;
```

Patients over 65:

```
1 SELECT medical_condition,
2         COUNT(*) AS number_of_patients_over_65
3 FROM Patient
4 WHERE age > 65
5 GROUP BY medical_condition;
```

Minimum & Maximum Age:

```
1 -- Min/Max age:
2 SELECT MIN(age) AS min_age, MAX(age) AS max_age
3 FROM Patient;
```

Oldest to Youngest Patients:

```
1 -- Oldest Patients to youngest:
2 SELECT patient_name,
3         age
4 FROM Patient
5 ORDER BY age DESC;
```

Insurance Check:

```
1 -- Insurance Check
2 SELECT DISTINCT insurance_provider
3 FROM Billing;
```

Top 10 Oldest Patient Information:

```
1 -- Top 10 oldest patient information:
2 SELECT TOP 10 patient_name,
3             medical_condition,
4             age
5 FROM Patient
6 ORDER BY age DESC;
```

Top 10 Youngest Patient Information:

```
1 -- Bottom 10 youngest patient information:
2 SELECT TOP 10 patient_name,
3             medical_condition,
4             age
5 FROM Patient
6 ORDER BY age;
```

Patient Blood Types Grouped:

```
1 -- Blood Types + Patient Count:
2 SELECT blood_type,
3        COUNT(*) as Patient_Count
4 FROM Patient
5 GROUP BY blood_type
6 ORDER BY Patient_Count DESC;
```

Average Age Between Genders:

```
1 SELECT AVG(age) as AverageAge, gender
2 FROM Patient
3 GROUP BY gender
4 ORDER BY AverageAge DESC;
```

Age Count + Condition Count:

```
1 -- Age Count + Condition Count:
2 SELECT
3     CASE
4         WHEN age BETWEEN 10 AND 20 THEN '10-20'
5         WHEN age BETWEEN 21 AND 30 THEN '21-30'
6         WHEN age BETWEEN 31 AND 40 THEN '31-40'
7         WHEN age BETWEEN 41 AND 50 THEN '41-50'
8         WHEN age BETWEEN 51 AND 60 THEN '51-60'
9         WHEN age BETWEEN 61 AND 70 THEN '61-70'
10        WHEN age BETWEEN 71 AND 80 THEN '71-80'
```

```

11         ELSE '81+'
12     END AS age_group,
13     medical_condition,
14     COUNT(*) AS total_patients
15 FROM Patient
16 GROUP BY
17     CASE
18         WHEN age BETWEEN 10 AND 20 THEN '10-20'
19         WHEN age BETWEEN 21 AND 30 THEN '21-30'
20         WHEN age BETWEEN 31 AND 40 THEN '31-40'
21         WHEN age BETWEEN 41 AND 50 THEN '41-50'
22         WHEN age BETWEEN 51 AND 60 THEN '51-60'
23         WHEN age BETWEEN 61 AND 70 THEN '61-70'
24         WHEN age BETWEEN 71 AND 80 THEN '71-80'
25         ELSE '81+'
26     END,
27     medical_condition
28 ORDER BY age_group, total_patients DESC;

```

Patients Per Hospital:

```

1 -- Patients per Hospital:
2 SELECT
3     d.hospital_name,
4     COUNT(p.patient_id) AS total_patients
5 FROM Patient p
6 JOIN Doctor d ON p.doctor_id = d.doctor_id
7 GROUP BY d.hospital_name
8 ORDER BY total_patients DESC;

```

Average Days by Condition:

```

1 -- Avg Treatment Days by Condition:
2 SELECT
3     p.medical_condition,
4     AVG(a.treatment_days) AS average_days
5 FROM Patient p
6 JOIN Admission a ON p.patient_id = a.patient_id
7 GROUP BY p.medical_condition
8 ORDER BY average_days DESC;

```

Total Billing by Insurance Provider:

```

1 -- Total Billing per Insurance Provider:
2 SELECT
3     b.insurance_provider,
4     SUM(a.billing_amount) AS total_bill

```

```

5 FROM Billing b
6 JOIN Patient p ON b.insurance_provider = p.insurance_provider
7 JOIN Admission a ON p.patient_id = a.patient_id
8 GROUP BY b.insurance_provider
9 ORDER BY total_bill DESC;

```

Most Common Conditions per Hospital:

```

1 -- Most Common Conditions per Hospital:
2 SELECT
3     d.hospital_name,
4     p.medical_condition,
5     COUNT(*) as case_count
6 FROM Patient p
7 JOIN Doctor d ON p.doctor_id = d.doctor_id
8 GROUP BY d.hospital_name, p.medical_condition
9 ORDER BY d.hospital_name, case_count DESC;

```

Top 10 Doctors with the Most Patients:

```

1 -- Top 10 Doctors with the most Patients:
2 SELECT TOP 10
3     d.doctor_name,
4     COUNT(p.patient_id) as total_patients
5 FROM Patient p
6 JOIN Doctor d on p.doctor_id = d.doctor_id
7 GROUP BY d.doctor_name
8 ORDER BY total_patients DESC;

```

Total Billing by Blood Type:

```

1 -- Total Billing by Blood Type:
2 SELECT
3     p.blood_type,
4     SUM(a.billing_amount) AS total_bill
5 FROM Patient p
6 JOIN Admission a ON p.patient_id = a.patient_id
7 GROUP BY p.blood_type
8 ORDER BY total_bill DESC;

```

Average Treatment Days by Admission Type:

```

1 -- Avg Treatment Days by Admission Type:
2 SELECT
3     admission_type,
4     AVG(treatment_days) AS avg_days

```

```

5 FROM Admission
6 GROUP BY admission_type
7 ORDER BY avg_days DESC;

```

Find the average billing amount for patients who had long treatments (over 15 days)

```

1 -- Find avg billing amt for patients who stayed longer than 15
  days:
2 WITH LongTreatments AS (
3     SELECT
4         p.patient_id,
5         a.treatment_days,
6         a.billing_amount
7     FROM Patient p
8     JOIN Admission a ON p.patient_id = a.patient_id
9     WHERE a.treatment_days >= 15
10 )
11 SELECT
12     CAST(ROUND(AVG(CAST(billing_amount AS DECIMAL(10,2))), 2) AS
13     DECIMAL(10,2)) AS avg_billing_for_long_treatments
14 FROM LongTreatments;

```

Total Billing Amount Per Insurance Provider by Blood Type

```

1 -- total billing amount per insurance provider by blood type:
2 WITH BillingDetails AS (
3     SELECT
4         p.blood_type,
5         p.insurance_provider,
6         a.billing_amount
7     FROM Patient p
8     JOIN Admission a ON p.patient_id = a.patient_id
9 )
10 SELECT
11     insurance_provider,
12     blood_type,
13     ROUND(SUM(billing_amount), 2) AS total_billed
14 FROM BillingDetails
15 GROUP BY insurance_provider, blood_type
16 ORDER BY insurance_provider, total_billed DESC;

```

Average Billing per Gender, For Patients with Emergency Admissions and Stayed for more than 5 Days:

```

1 -- Average billing amount per gender, but only for patients who
  had emergency admissions and stayed more than 5 days

```



```

2 WITH EmergencyStays AS (
3     SELECT
4         p.gender,
5         a.billing_amount
6     FROM Patient p
7     JOIN Admission a ON p.patient_id = a.patient_id
8     WHERE a.admission_type = 'EMERGENCY' AND a.treatment_days > 5
9 )
10 SELECT
11     gender,
12     CAST(ROUND(AVG(billing_amount), 2) AS FLOAT) AS
13     avg_emergency_billing
14 FROM EmergencyStays
15 GROUP BY gender
16 ORDER BY avg_emergency_billing DESC;

```

Top 3 Admissions by Total Billing For Patients with Blood Type 'O'

```

1 -- Top 3 Admissions by Total Billing For Patients with Blood
  Type 'O'
2 WITH BloodOAdmissions AS (
3     SELECT
4         a.admission_type,
5         a.billing_amount
6     FROM Patient p
7     JOIN Admission a ON p.patient_id = a.patient_id
8     WHERE p.blood_type LIKE 'O%'
9 ),
10
11 BillingType AS (
12     SELECT
13         admission_type,
14         SUM(billing_amount) AS total_billed
15 FROM BloodOAdmissions
16 GROUP BY admission_type
17 )
18 SELECT
19     admission_type,
20     ROUND(total_billed, 2) AS total_billed
21 FROM BillingType
22 ORDER BY total_billed DESC;

```

Doctor's Total Number of Patients with Average Treatment Duration

```

1 -- Doctors total number of patients and their average
  treatment duration

```

```

2 WITH DoctorAdmissions AS (
3     SELECT
4         d.doctor_id,
5         d.doctor_name,
6         a.treatment_days
7     FROM Doctor d
8     JOIN Patient p ON d.doctor_id = p.doctor_id
9     JOIN Admission a ON p.patient_id = a.patient_id
10 ),
11 DoctorStats AS (
12     SELECT
13         doctor_id,
14         doctor_name,
15         COUNT(*) AS total_patients,
16         ROUND(AVG(treatment_days), 2) AS avg_treatment_days
17     FROM DoctorAdmissions
18     GROUP BY doctor_id, doctor_name
19 )
20 SELECT
21     doctor_name,
22     total_patients,
23     avg_treatment_days
24 FROM DoctorStats
25 ORDER BY total_patients DESC;

```
